Year 12 Maths - Applications of differentiation

Similar questions - Stationary points and their nature

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A curve has equation $y = x^3 - 3x^2 - 9x$.

(a) Find the coordinates of the stationary points of the curve and determine their nature. **[6]**

(b) The equation $x^3 - 3x^2 - 9x + k = 0$, where k is a constant, has three distinct real roots. Find the set of possible values of k. **[3]**

2 A curve has equation $y = x^2 + \dfrac{54}{x}$ for $x \neq 0$.

(a) Find the coordinates of the stationary point and use the second derivative to determine its nature. **[6]**

(b) Find the set of values of x for which y is increasing. **[2]**

(c) Hence explain why the curve has no maximum point. **[1]**

Question 1

(a) $\dfrac{dy}{dx} = 3x^2 - 6x - 9 = 3(x-3)(x+1)$, zero at $x = -1, 3$. Points $(-1, 5)$ and $(3, -27)$. $\dfrac{d^2y}{dx^2} = 6x - 6$: at $x = -1$ it is $-12 < 0$ (maximum); at $x = 3$ it is $12 > 0$ (minimum)

(b) Three distinct roots need the curve $y = x^3 - 3x^2 - 9x$ to meet the line $y = -k$ three times, so $-27 < -k < 5$. Hence $-5 < k < 27$

Question 2

(a) $\dfrac{dy}{dx} = 2x - 54x^{-2} = 0$ gives $x^3 = 27$, so $x = 3$ and $y = 9 + 18 = 27$. Point $(3, 27)$. $\dfrac{d^2y}{dx^2} = 2 + 108x^{-3} = 6 > 0$, so a minimum

(b) $\dfrac{dy}{dx} = \dfrac{2x^3 - 54}{x^2} > 0$ when $x^3 > 27$, i.e. $x > 3$

(c) The only stationary point is the minimum; the gradient is negative on both sides of the y-axis until $x = 3$, and positive after, so the curve never turns downwards after a rise (no maximum)